CISC 6090: Capstone in Cybersecurity

Instructor: Dr. Thaier Hayajneh, University Professor & Director
Semester: Fall 2026
Credit: 3 hours
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Office Hours: by appointment.

Course Objectives:
The goal of this class is to sharpen students' skills in cybersecurity by doing a capstone project. After this class, students should deeply understand state-of-the-art cybersecurity technologies and knowledge. Students are required to finish **a significant capstone project paper** in this class.

Learning Outcomes:
The learning outcomes of this course are:
1. An ability to apply knowledge obtained in earlier coursework and obtain new knowledge necessary to design, test, and verify/validate a system, program, protocol, component, or process to meet the desired needs of cybersecurity
2. An understanding of the engineering design and programming process related to cybersecurity
3. An ability to function on a multidisciplinary team
4. An awareness of professional and ethical responsibility
5. An ability to communicate effectively in both oral and written form.

Textbook:
None

Recommended References:
Conference and journal papers.

Grading Scheme
The overall grading scheme will be based on the following criteria:

Presentations:	30%
Capstone project:	70%
Total:	**100%**

Grading of Capstone project will be based on the following criteria:

Originality and Technical Content:	15 Points
Results and Discussion:	20 Points
Documentation:	15 Points
Presentation:	10 Points
Paper Quality:	10 points
Total Capstone Project Grade:	**70 Points**

Course Policy

- Each student or group should finish their required project assignment on time. Submissions must be on time to be accepted.
- The student must check the Blackboard system regularly to get related course information.
- Copying material from the Web/Internet or other project sources will be considered cheating.

- Each student should submit a well-documented final (capstone) project paper with proper sections. Typically, the paper should have the following sections: a title, name of student and e-mail address, abstract, introduction, literature review, problem statement, methodology, results and discussion, conclusion, and list of references. At the end, the source code has to be listed to indicate how it can be run and on which platform.

Course Web Site

- The course homepage can be accessed through Blackboard in your ***my.Fordham*** Account.
- All course materials and announcements will be posted there.

Miscellaneous

1. Please let your instructor know if you have a disability that requires special arrangements/accommodation. Under the Americans with Disabilities Act and Section 504 of the Vocational Rehabilitation Act of 1973, all students, with or without disabilities, are entitled to equal access to the programs and activities of Fordham University. If you believe that you have a disability that may interfere with your ability to participate in the activities, coursework, or assessment of the of this course, you may be entitled to accommodations. Please schedule a meeting to speak with someone at the Office of Disability Services (Rose Hill - O'Hare Hall, Lower Level, x0655 or at Lincoln Center – Room 207, x6282).
2. Students agree that, by taking this course, all required papers may be subject to submission for textual similarity review for the detection of plagiarism. All submitted papers will be included as source documents in the plagiarism detection software reference database solely for the purpose of detecting plagiarism of such papers.

Class schedule

Class	Topic
1 (08/29/2026)	Introduction and course syllabus, project description, review of sample papers and projects, Dr. Hayajneh Research presentation
2 (09/19/2026)	Project proposal presentation and paper presentation I
3 (10/03/2026)	Related paper presentation I
4 (10/17/2026)	Related work presentation II
5 (10/31/2026)	Project progress report presentation
6 (11/14/2026)	Open discussion
7 (12/05/2026)	Final project presentation

Capstone project

Project Overview

In this Project and Internship course project, student(s) will identify and investigate problems of current interest in cybersecurity research. Each group of students will choose a topic within the scope of the Cybersecurity program, develop and propose a research hypothesis, and work toward solving the problem of interest. Students are required to agree on a project topic with the instructor before developing a project proposal. Depending on the chosen topic, teams will develop novel techniques toward the desired goal, implement their approach in a simulation/emulation environment or in an appropriate system (e.g., mobile phones, sensor nodes, SDR kits), and evaluate the performance and security impacts of their approaches using appropriate metrics.

Project Requirements

Teams of 1-2 students will be formed during the first week of class. Under the guidance of the instructor, each team will choose a project topic by the third week of the semester, submit a written project proposal and present it to the class. Each team will present a progress report to the class mid-way through the semester, and a final project presentation will be made during the last week of class. In addition, each team will submit a written project report, including documentation of any hardware or software systems developed as part of the project, at the end of the semester. In certain cases, the final project report can be replaced by a conference or workshop paper, poster, or demo submission. The scope and depth of the project must be appropriate for a semester-long course project, and project components can be added, removed, or modified throughout the semester.

Deliverables

The initial project proposal has two parts: 1) a written proposal, not to exceed 2 pages in length, that provides an overview of the project and the expected outcomes and 2) a 10-minute in-class presentation on the specified day. Project teams will present a 15-minute in-class progress report on the specified day, including the following 1) an overview of the project, 2) milestones achieved to date, 3) roadblocks faced and how they affect the project goals, and 4) a detailed time-line of the team's efforts; note that the progress report does not have a written component.
Related work section: This is usually a 1 or 2 page in which you discuss others' work (papers) that are related to your paper. You should at least include 4-5 papers (the more the better!), discuss the papers briefly, compare them and finally highlight how is your paper going to be different. The sample papers, the papers you presented and collected are all good examples and include a similar section.
The final project deliverable at the end of the semester has two parts: 1) a written report, make sure to follow the required format (in IEEE Journal format), including an overview, background, related work, results, and documentation and 2) and in-class presentation, including a demo if applicable.

Final project paper format and structure, your paper should:

- Be at least 5000 words. There is no upper limit.
- Have at least 10-15 references with at least 10 of those being research papers (conferences and journals).
- All references must be cited properly in your paper.
- Have a structure similar to the sample papers and the papers you presented in class, this includes Abstract (150-250 words), Introduction (1-2 pages), background about the topic or literature review section, A section for your proposed problem/ system/ comparison metric/ idea…etc, a result section, maybe a discussion section, finally a conclusion section (not more than half a page).
- All figures and tables must have captions and described in the paper.
- I recommend presenting your data/ results in the form of figures, charts, or tables…

Blackboard has two templates that you can use for your final paper.

Deliverable Time-Line

The following schedule of project deliverables has been tentatively set and is subject to changes.

- Project Proposal: Sep. 19, 2026
- Related work: Oct. 17, 2026
- Progress Report: Oct. 31 , 2026
- Final Presentation: Dec. 5 , 2026
- Project Report: Dec. 3 , 2026 (Draft), Dec. 09, 2026 (Final).